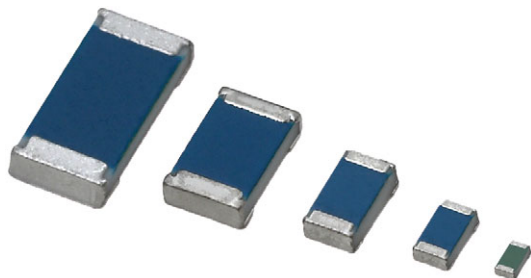


High Stability Thin Film Flat Chip Resistors



FEATURES

- Rated dissipation P_{70} up to 0.52 W for size 1206
- AEC-Q200 qualified
- Products are PFAS free
- IECQ-CECC approved to EN 140401-801
- Operating temperature up to 175 °C
- Superior temperature cycling robustness
- Superior moisture resistivity, $|\Delta R/R| < 0.1 \%$ (85 °C; 85 % RH; 1000 h)
- Superior sulfur resistance verified according to ASTM B 809 and ANSI/EIA-977
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



LINKS TO ADDITIONAL RESOURCES



3D Models



Product Page



Infographics



Order Samples

Vishay Automotive Grade MC AT thin film chip resistors are the perfect choice for most fields of modern electronics where reliability and stability is of major concern. Typical applications include automotive, industrial, telecommunication and medical equipment, as well as precision test and measuring equipment.

APPLICATIONS

- Automotive
- Industrial equipment with enhanced requirements
- Telecommunication
- Medical equipment

TECHNICAL SPECIFICATIONS					
DESCRIPTION	MCR 0201 AT	MCS 0402 AT	MCT 0603 AT	MCU 0805 AT	MCA 1206 AT
Imperial size	0201	0402	0603	0805	1206
Metric size code	RR0603M	RR1005M	RR1608M	RR2012M	RR3216M
Resistance range	22 Ω to 40 k Ω	1 Ω to 1 M Ω ; 0 Ω	1 Ω to 2 M Ω ; 0 Ω	1 Ω to 7.5 M Ω ; 0 Ω	1 Ω to 10 M Ω ; 0 Ω
Resistance tolerance	$\pm 0.5 \%$; $\pm 0.1 \%$	$\pm 1 \%$; $\pm 0.5 \%$; $\pm 0.1 \%$	$\pm 1 \%$; $\pm 0.5 \%$; $\pm 0.1 \%$; $\pm 0.05 \%$	$\pm 1 \%$; $\pm 0.5 \%$; $\pm 0.1 \%$	
Temperature coefficient	± 25 ppm/K	± 50 ppm/K; ± 25 ppm/K; ± 15 ppm/K; ± 10 ppm/K	± 50 ppm/K; ± 25 ppm/K; ± 15 ppm/K; ± 10 ppm/K; ± 5 ppm/K	± 50 ppm/K; ± 25 ppm/K; ± 15 ppm/K; ± 10 ppm/K	
Rated dissipation P_{70} ⁽¹⁾	0.075 W	0.130 W	0.210 W	0.260 W	0.520 W
Rated dissipation, P at terminal part temperature, T_A ⁽²⁾	0.25 W	0.33 W	0.5 W	0.75 W	1 W
Operating voltage, U_{max} , AC _{RMS} /DC	25 V	75 V	100 V	150 V	200 V
Permissible film temperature, T_F max. ⁽¹⁾	175 °C				
Operating temperature range ⁽³⁾	-55 °C to 175 °C				
Permissible voltage against ambient (insulation): 1 min; U_{ins}	50 V	75 V	100 V	200 V	300 V
Failure rate: FIT _{observed}	$\leq 0.1 \times 10^{-9}/h$				

Notes

- ⁽¹⁾ Please refer to APPLICATION INFORMATION, see below
- ⁽²⁾ The specified maximum terminal part temperature must not be exceeded; compliance shall be ensured by appropriate thermal management and PCB design at the customer's application level
- ⁽³⁾ Please refer to table MAXIMUM RESISTANCE CHANGE AT RATED DISSIPATION, see below

APPLICATION INFORMATION

When the resistor dissipates power, a temperature rise above the ambient temperature occurs, dependent on the thermal resistance of the assembled resistor together with the printed circuit board. The rated dissipation applies only if the permitted film temperature is not exceeded.

Please consider the application note "Thermal Management in Surface-Mounted Resistor Applications" (www.vishay.com/doc?28844) for information on the general nature of thermal resistance.

These resistors do not feature a limited lifetime when operated within the permissible limits. However, resistance value drift increasing over operating time may result in exceeding a limit acceptable to the specific application, thereby establishing a functional lifetime.



MAXIMUM RESISTANCE CHANGE AT RATED DISSIPATION							
OPERATION MODE		GENERAL		POWER		ADVANCED	
Rated dissipation, P_{70}	MCR 0201 AT	0.050 W		0.075 W		-	
	MCS 0402 AT	0.070 W		0.105 W		0.130 W	
	MCT 0603 AT	0.110 W		0.170 W		0.210 W	
	MCU 0805 AT	0.140 W		0.220 W		0.260 W	
	MCA 1206 AT	0.270 W		0.420 W		0.520 W	
Operating temperature range		-55 °C to 125 °C		-55 °C to 155 °C		-55 °C to 175 °C	
Permissible film temperature, ϑ_F max.		125 °C		155 °C		175 °C	
Max. resistance change at P_{70} for resistance range, $ \Delta R/R $ after:	MCR 0201 AT	22 Ω to 40 k Ω	-	22 Ω to 40 k Ω	-	-	-
	MCS 0402 AT	1 Ω to $\leq$ 47 k Ω	> 47 k Ω to 1 M Ω	1 Ω to $\leq$ 47 k Ω	> 47 k Ω to 1 M Ω	1 Ω to $\leq$ 47 k Ω	> 47 k Ω to 1 M Ω
	MCT 0603 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 2 M Ω	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 2 M Ω	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 2 M Ω
	MCU 0805 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 7.5 M Ω	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 7.5 M Ω	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 7.5 M Ω
	MCA 1206 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 10 M Ω	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 10 M Ω	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 10 M Ω
	1000 h	$\leq$ 0.05 %	$\leq$ 0.1 %	$\leq$ 0.1 %	$\leq$ 0.2 %	$\leq$ 0.2 %	$\leq$ 0.5 %
	8000 h	$\leq$ 0.1 %	$\leq$ 0.2 %	$\leq$ 0.2 %	$\leq$ 0.4 %	-	-
	225 000 h ⁽¹⁾	$\leq$ 0.3 %	$\leq$ 0.6 %	$\leq$ 0.6 %	-	-	-

Notes

- The presented operation modes do not refer to different types of resistors, but actually show examples of different loads, that lead to different film temperatures and different achievable load-life stability (drift) of the resistance value. A suitable low thermal resistance of the circuit board assembly must be safeguarded in order to maintain the film temperature of the resistors within the specified limits. Please consider the application note "Thermal Management in Surface-Mounted Resistor Applications" (www.vishay.com/doc?28844) for information on the general nature of thermal resistance

(1) Calculated

TEMPERATURE COEFFICIENT AND RESISTANCE RANGE ⁽¹⁾				
TYPE / SIZE	TCR	TOLERANCE	RESISTANCE	E-SERIES
MCR 0201 AT	$\pm$ 25 ppm/K	$\pm$ 0.5 % $\pm$ 0.1 %	22 Ω to 40 k Ω 22 Ω to 25 k Ω	E24; E192
MCS 0402 AT	$\pm$ 50 ppm/K	$\pm$ 1 %	1 Ω to 221 k Ω	E24; E96
	$\pm$ 25 ppm/K	$\pm$ 0.5 %	10 Ω to 221 k Ω	E24; E192
	$\pm$ 25 ppm/K	$\pm$ 0.1 %	47 Ω to 1 M Ω	E24; E192
	$\pm$ 15 ppm/K		47 Ω to 100 k Ω	
	$\pm$ 10 ppm/K		47 Ω to 47 k Ω	
	Jumper ⁽²⁾ ; $I_{max.} = 0.63$ A	$\leq$ 20 m Ω	0 Ω	-
MCT 0603 AT	$\pm$ 50 ppm/K	$\pm$ 1 %	1 Ω to 511 k Ω	E24; E96
	$\pm$ 25 ppm/K	$\pm$ 0.5 %	10 Ω to 511 k Ω	E24; E192
	$\pm$ 25 ppm/K	$\pm$ 0.1 %	47 Ω to 2 M Ω	E24; E192
	$\pm$ 15 ppm/K		47 Ω to 221 k Ω	
	$\pm$ 10 ppm/K		47 Ω to 100 k Ω	
	$\pm$ 5 ppm/K		3.9 k Ω to 100 k Ω	
	$\pm$ 5 ppm/K	$\pm$ 0.05 %	3.9 k Ω to 100 k Ω	
	Jumper ⁽²⁾ ; $I_{max.} = 1$ A	$\leq$ 20 m Ω	0 Ω	-
MCU 0805 AT	$\pm$ 50 ppm/K	$\pm$ 1 %	1 Ω to 1 M Ω	E24; E96
	$\pm$ 25 ppm/K	$\pm$ 0.5 %	10 Ω to 1 M Ω	E24; E192
	$\pm$ 25 ppm/K	$\pm$ 0.1 %	47 Ω to 7.5 M Ω	E24; E192
	$\pm$ 15 ppm/K		47 Ω to 332 k Ω	
	$\pm$ 10 ppm/K		47 Ω to 100 k Ω	
	Jumper ⁽²⁾ ; $I_{max.} = 1.5$ A	$\leq$ 20 m Ω	0 Ω	-
MCA 1206 AT	$\pm$ 50 ppm/K	$\pm$ 1 %	1 Ω to 1 M Ω	E24; E96
	$\pm$ 25 ppm/K	$\pm$ 0.5 %	10 Ω to 1 M Ω	E24; E192
	$\pm$ 25 ppm/K	$\pm$ 0.1 %	47 Ω to 10 M Ω	E24; E192
	$\pm$ 15 ppm/K		47 Ω to 511 k Ω	
	$\pm$ 10 ppm/K		47 Ω to 100 k Ω	
	Jumper ⁽²⁾ ; $I_{max.} = 2$ A	$\leq$ 20 m Ω	0 Ω	-

Notes(1) For the approved IECQ-CECC resistance range, including stability class information, please refer to www.vishay.com/doc?28945(2) The temperature coefficient of resistance (TCR) is not specified for 0 Ω jumpers

**PACKAGING**

TYPE / SIZE	CODE	QUANTITY	PACKAGING STYLE	WIDTH	PITCH	PACKAGING DIMENSIONS
MCR 0201 AT, MCS 0402 AT	E0	10 000	Paper tape acc. IEC 60286-3, Type 1a	8 mm	2 mm	Ø 180 mm / 7"
MCT 0603 AT, MCU 0805 AT	P5	5000			4 mm	Ø 330 mm / 13"
	PW	20 000				Ø 180 mm / 7"
MCA 1206 AT	P5	5000				

PART NUMBER AND PRODUCT DESCRIPTION

Part Number: MCT0603MD4641DPW00

Part Number: MCT0603MZ0000ZP500

M	C	T	0	6	0	3	M	D	4	6	4	1	D	P	W	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

TYPE / SIZE	VERSION	TCR	RESISTANCE	TOLERANCE	PACKAGING	SPECIAL
MCR0201 MCS0402 MCT0603 MCU0805 MCA1206	M = AT (automotive)	G = ± 5 ppm/K F = ± 10 ppm/K E = ± 15 ppm/K D = ± 25 ppm/K C = ± 50 ppm/K Z = jumper	3 digit value 1 digit multiplier Multiplier 8 = *10 ⁻² 9 = *10 ⁻¹ 0 = *10 ⁰ 1 = *10 ¹ 2 = *10 ² 3 = *10 ³ 4 = *10 ⁴ 5 = *10 ⁵	A = ± 0.05 % B = ± 0.1 % D = ± 0.5 % F = ± 1 % Z = jumper	E0 P5 PW	Up to 2 digits 00 = standard

Product Description: MCT 0603-25 0.5 % AT PW 4K64

Product Description: MCT 0603 AT P5 0R0

MCT	0603	-25	0.5 %	AT	PW	4K64
TYPE	SIZE	TCR	TOLERANCE	VERSION	PACKAGING	RESISTANCE
MCR MCS MCT MCU MCA	0201 0402 0603 0805 1206	± 5 ppm/K ± 10 ppm/K ± 15 ppm/K ± 25 ppm/K ± 50 ppm/K	± 0.05 % ± 0.1 % ± 0.5 % ± 1 %	AT = automotive	E0 P5 PW	4K64 = 4.64 kΩ 0R0 = jumper

Note

- Products can be ordered using either the PART NUMBER or PRODUCT DESCRIPTION

**DESCRIPTION**

Production is strictly controlled and follows an extensive set of instructions established for reproducibility. A homogeneous film of special metal alloy is deposited on a high grade ceramic substrate (Al_2O_3) and conditioned to achieve the desired temperature coefficient. Specially designed inner contacts are deposited on both sides. A special laser is used to achieve the target value by smoothly cutting a meander groove in the resistive layer without damaging the ceramics. The resistor elements are covered by a unique protective coating designed for electrical, mechanical and climatic protection. The terminations receive a final pure matte tin on nickel plating.

The result of the determined production is verified by an extensive testing procedure and optical inspection performed on 100 % of the individual chip resistors. This includes full screening for the elimination of products with potential risk of early field failures (feasible for $R \geq 10 \Omega$). Only accepted products are laid directly into the paper tape in accordance with **IEC 60286-3, Type 1a** ⁽¹⁾.

ASSEMBLY

The resistors are suitable for processing on automatic SMD assembly systems. They are suitable for automatic soldering using wave, reflow or vapour phase as shown in **IEC 61760-1** ⁽¹⁾. The encapsulation is resistant to all cleaning solvents commonly used in the electronics industry, including alcohols, esters and aqueous solutions. The suitability of conformal coatings, potting compounds and their processes, if applied, shall be qualified by appropriate means to ensure the long-term stability of the whole system.

The resistors are RoHS-compliant, the pure matte tin plating provides compatibility with lead (Pb)-free and lead-containing soldering processes. Solderability is specified for 2 years after production or requalification. The permitted storage time is 20 years. The immunity of the plating against tin whisker growth has been proven under extensive testing.

MATERIALS

Vishay acknowledges the following systems for the regulation of hazardous substances:

- IEC 62474, Material Declaration for Products of and for the Electrotechnical Industry, with the list of declarable substances given therein ⁽²⁾
- The Global Automotive Declarable Substance List (GADSL) ⁽³⁾
- The REACH regulation (1907/2006/EC) and the related list of substances with very high concern (SVHC) ⁽⁴⁾ for its supply chain

Notes

- ⁽¹⁾ The quoted IEC standards are also released as EN standards with the same number and identical contents
- ⁽²⁾ The IEC 62474 list of declarable substances is maintained in a dedicated database, which is available at <http://std.iec.ch/iec62474>
- ⁽³⁾ The Global Automotive Declarable Substance List (GADSL) is maintained by the American Chemistry Council and available at www.gadsl.org
- ⁽⁴⁾ The SVHC list is maintained by the European Chemical Agency (ECHA) and available at <http://echa.europa.eu/candidate-list-table>
- ⁽⁵⁾ For the approved IECQ-CECC resistance range, please refer to www.vishay.com/doc?28945

The products do not contain any of the banned substances as per IEC 62474, GADSL, or the SVHC list, see www.vishay.com/how/leadfree.

Hence the products fully comply with the following directives:

- 2000/53/EC End-of-Life Vehicle Directive (ELV) and Annex II (ELV II)
- 2011/65/EU Restriction of the Use of Hazardous Substances Directive (RoHS) with amendment 2015/863/EU
- 2012/19/EU Waste Electrical and Electronic Equipment Directive (WEEE)

Vishay pursues the elimination of conflict minerals from its supply chain, see the Conflict Minerals Policy at www.vishay.com/doc?49037.

APPROVALS

Where applicable, the resistors are approved within the IECQ-CECC Quality Assessment System for Electronic Components to the detail specification **EN 140401-801** which refers to **EN 60115-1**, **EN 60115-8** and the variety of environmental test procedures of the **IEC 60068** ⁽¹⁾ series.

Conformity is attested by the use of the **CECC** ⁽⁵⁾ logo () as the mark of conformity on the package label.

Vishay Beyschlag has achieved “**Approval of Manufacturer**” in accordance with **IECQ 03-1**. The release certificate for “**Technology Approval Schedule**” in accordance with **CECC 240001** based on **IECQ 03-3-1** is granted for the Vishay Beyschlag manufacturing process.

The resistors are qualified according to AEC-Q200.

RELATED PRODUCTS

Chip resistor arrays may be used in sensing applications or precision amplifiers where close matching between multiple resistors is necessary. Please refer to the **ACAS AT - Precision** datasheet (www.vishay.com/doc?28770).

MC AT series is also available with gold termination for conductive gluing: **MC ATAU - Precision**. Please refer to the datasheet (www.vishay.com/doc?28877).

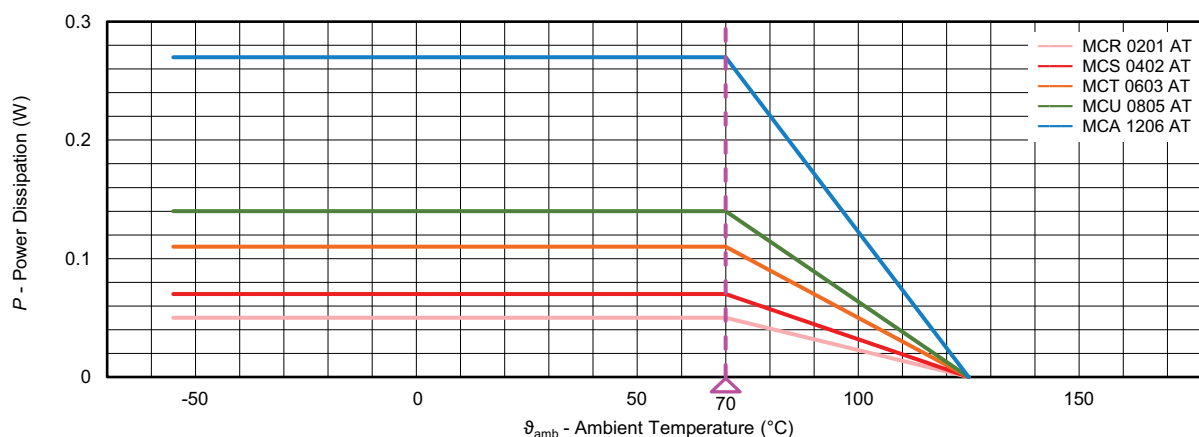
For high power and high temperature applications **MCW AT** wide terminal thin film chip resistors offer extremely high power ratings and extraordinary temperature cycling robustness. Please refer to the datasheets: www.vishay.com/doc?28847 and www.vishay.com/doc?28796.

These wide-terminal products are also available in low-ohmic values, **NCW AT** (www.vishay.com/doc?28849).

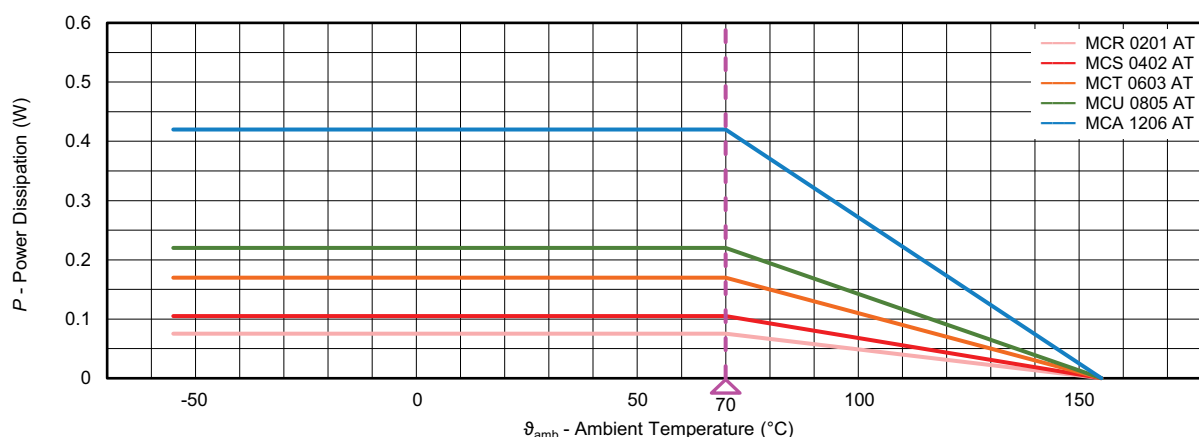


FUNCTIONAL PERFORMANCE

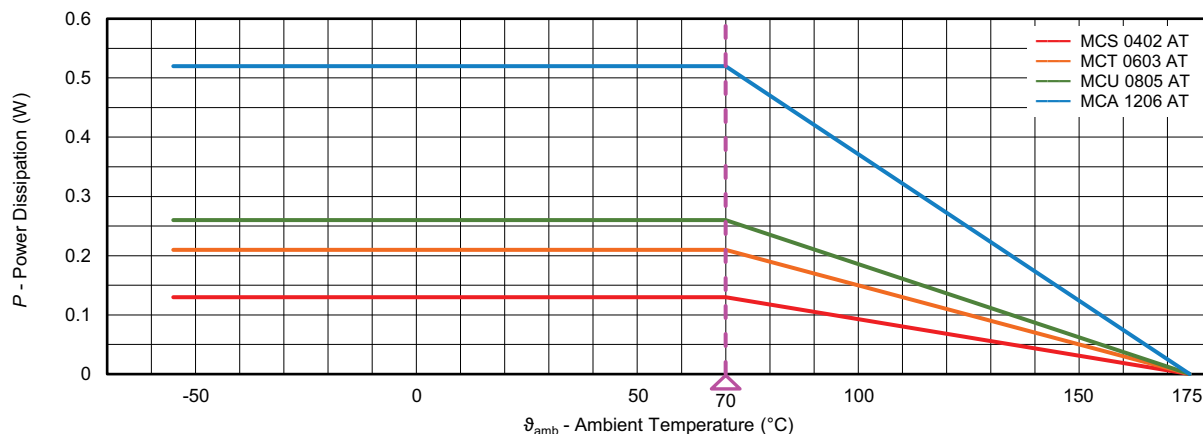
PERFORMANCE IN SULFUR-CONTAINING AMBIANCE		
TEST NAME	HUMID SULFUR VAPOR TEST FOS-90	DRY SULFUR TEST FOS-105
Reference specification	ASTM B 809	ANSI/EIA-977
Test conditions (temperature, humidity)	90 °C ± 2 °C 76 % to 95 % RH	105 °C ± 2 °C RH not specified, not applicable
Aggressive agent	Sulfur (saturated vapor)	Sulfur, sublimed
Failure criteria in VI under magnification	Silver sulfide growth: length shall be < 50 µm	Not applicable
Failure criteria in electrical test	$\leq (\pm 1 \% R + 0.05 \Omega)$	$\leq (\pm 1 \% R + 0.05 \Omega)$
Test duration	1000 h	1000 h



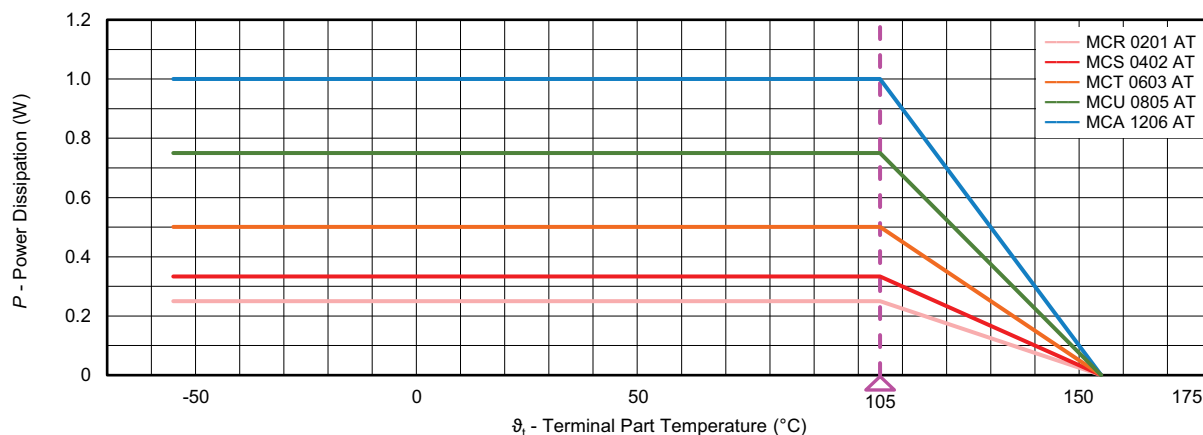
Derating - General Operation



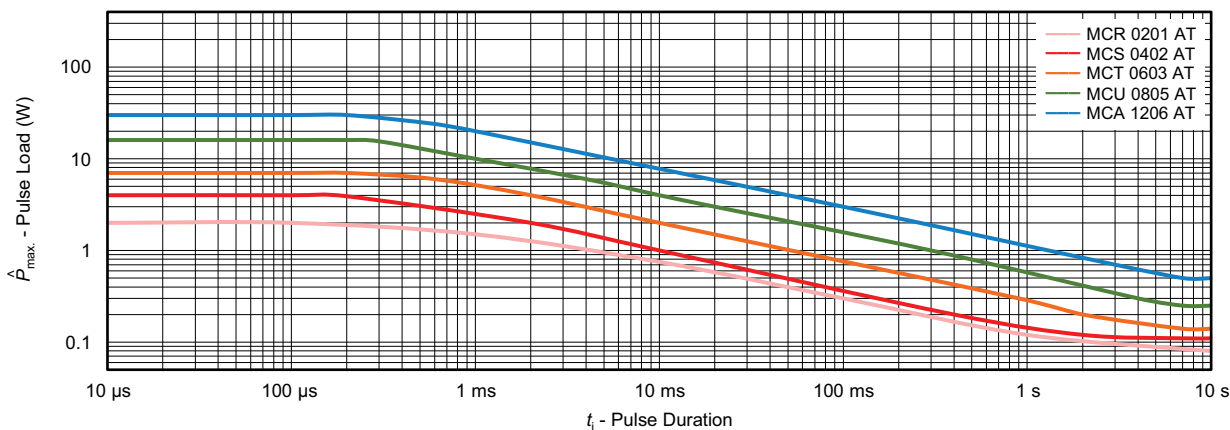
Derating - Power Operation



Derating - Advanced Operation

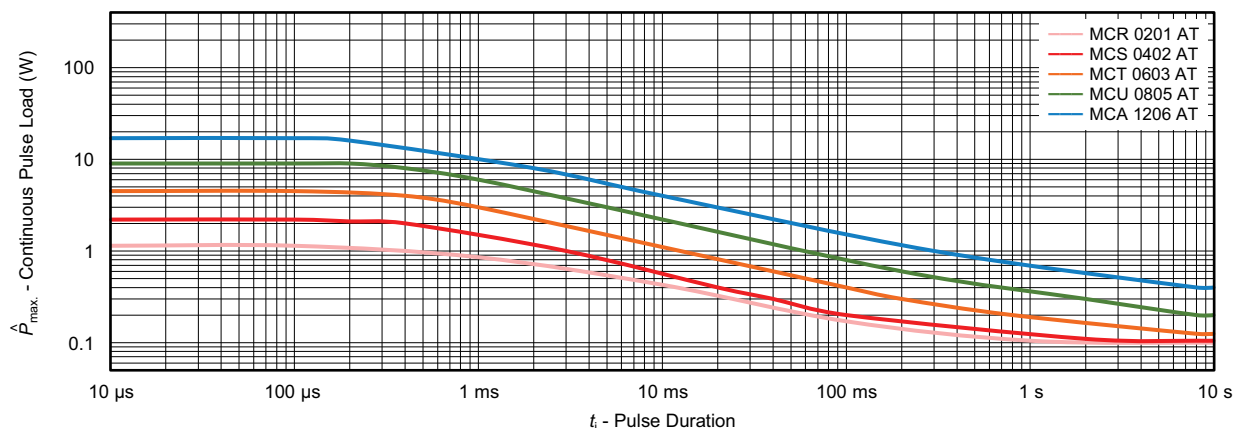


Derating - Terminal Part Temperature



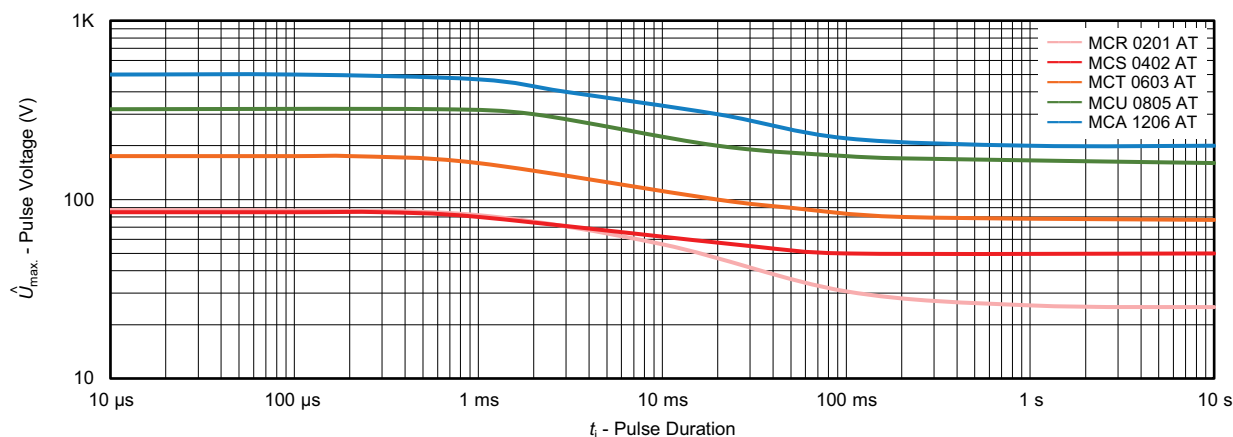
Maximum pulse load, single pulse; applicable if $\bar{P} \rightarrow 0$ and $n \leq 1000$ and $\dot{U} \leq \dot{U}_{\max}$;
for permissible resistance change $\pm (0.5 \% R + 0.05 \Omega)$

Single Pulse



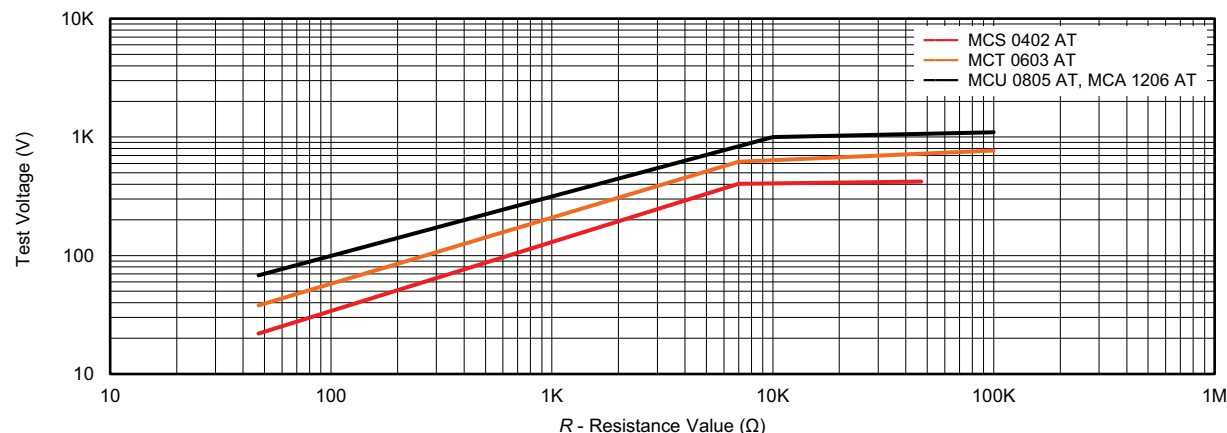
Maximum pulse load, continuous pulse; applicable if $\bar{P} \leq P(\vartheta_{amb})$ and $\hat{U} \leq \hat{U}_{max}$;
for permissible resistance change $\pm (0.5 \% R + 0.05 \Omega)$

Continuous Pulse



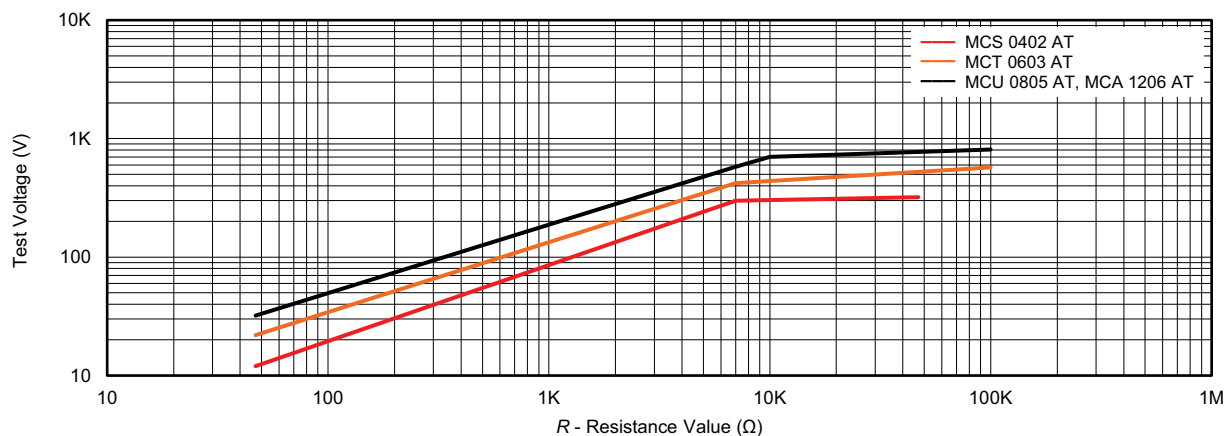
Maximum pulse voltage, single and continuous pulses; applicable if $\hat{P} \leq \hat{P}_{max}$;
for permissible resistance change $\pm (0.5 \% R + 0.05 \Omega)$

Pulse Voltage



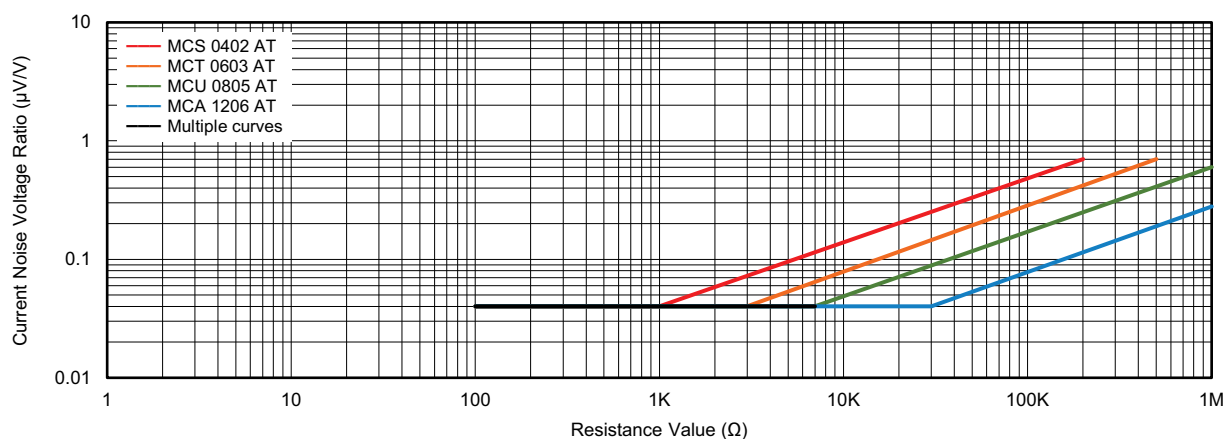
Pulse load rating in accordance with EN 60115-1 clause 4.27; 1.2 μ s/50 μ s;
5 pulses at 12 s intervals; for permissible resistance change $\pm (0.5 \% R + 0.05 \Omega)$

1.2/50 Pulse



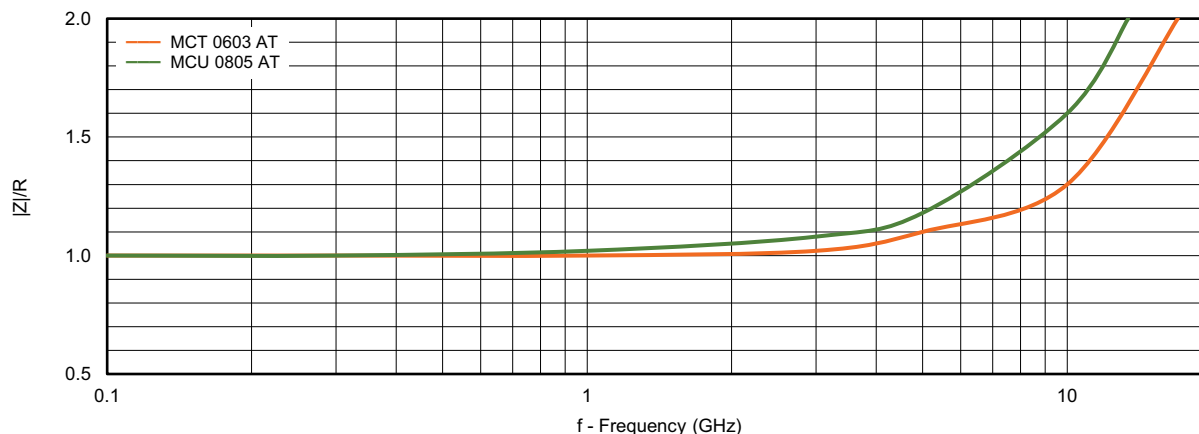
Pulse load rating in accordance with EN 60115-1 clause 4.27; 10 μ s/700 μ s;
10 pulses at 1 minute intervals; for permissible resistance change $\pm (0.5 \% R + 0.05 \Omega)$

10/700 Pulse



In accordance with IEC 60195

Current Noise Voltage Ratio



|Z|/R for 49.9 chip resistor

RF-Behavior

**TESTS AND REQUIREMENTS**

All tests are carried out in accordance with the following specifications:

EN 60115-1, generic specification

EN 60115-8, sectional specification

EN 140401-801, detail specification

IEC 60068-2-xx, test methods

When applicable, the components are approved under the IECQ-CECC quality assessment system for electronic components.

The parameters stated in the Test Procedures and Requirements table are based on the required tests and permitted limits of EN 140401-801. The table presents only the most important tests, for the full test schedule refer to the documents listed above. However, some additional tests and a number of improvements against those minimum requirements have been included.

The testing also covers most of the requirements specified by EIA/ECA-703 and JIS-C-5201-1.

The tests are carried out under standard atmospheric conditions in accordance with IEC 60068-1, 4.3, whereupon the following values are applied:

Temperature: 15 °C to 35 °C

Relative humidity: 25 % to 75 %

Air pressure: 86 kPa to 106 kPa (860 mbar to 1060 mbar)

A climatic category LCT / UCT / 56 is applied, defined by the lower category temperature (LCT), the upper category temperature (UCT), and the duration of exposure in the damp heat, steady state test (56 days).

The components are mounted for testing on printed circuit boards in accordance with EN 60115-8, 5.2.2, unless otherwise specified.

TEST PROCEDURES AND REQUIREMENTS					
EN 60115-1 CLAUSE	IEC 60068-2 ⁽¹⁾ TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR)	
				HIGH ACCURACY	REGULAR ACCURACY
			Stability for product types:		
			MCR 0201 AT	22 Ω to 40 k Ω	-
			MCS 0402 AT	1 Ω to $\leq$ 47 k Ω	> 47 k Ω to 1 M Ω
			MCT 0603 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 2 M Ω
			MCU 0805 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 7.5 M Ω
			MCA 1206 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 10 M Ω
6.1	-	Resistance		$\pm 1\ \% R$; $\pm 0.5\ \% R$; $\pm 0.1\ \% R$; $\pm 0.05\ \% R$	
6.2	-	Temperature coefficient	At (20 / -55 / 20) °C and (20 / 155 / 20) °C	$\pm 50\ \text{ppm/K}$; $\pm 25\ \text{ppm/K}$; $\pm 15\ \text{ppm/K}$; $\pm 10\ \text{ppm/K}$; $\pm 5\ \text{ppm/K}$	
7.1	-	Endurance at 70 °C: general operation mode	$U = \sqrt{P_{70} \times \overline{R}}$ or $U = U_{\text{max.}}$; whichever is the less severe; 1.5 h on; 0.5 h off; 70 °C; 1000 h 70 °C; 8000 h 70 °C; 225 000 h	$\pm (0.05\ \% R + 0.01\ \Omega)$ $\pm (0.10\ \% R + 0.02\ \Omega)$ $\pm (0.30\ \% R + 0.02\ \Omega)$	$\pm (0.10\ \% R + 0.02\ \Omega)$ $\pm (0.20\ \% R + 0.02\ \Omega)$ $\pm (0.60\ \% R + 0.05\ \Omega)$
		Endurance at 70 °C: power operation mode	$U = \sqrt{P_{70} \times \overline{R}}$ or $U = U_{\text{max.}}$; whichever is the less severe; 1.5 h on; 0.5 h off; 70 °C; 1000 h 70 °C; 8000 h 70 °C; 225 000 h	$\pm (0.10\ \% R + 0.01\ \Omega)$ $\pm (0.20\ \% R + 0.02\ \Omega)$ $\pm (0.60\ \% R + 0.05\ \Omega)$	$\pm (0.20\ \% R + 0.02\ \Omega)$ $\pm (0.40\ \% R + 0.05\ \Omega)$
		Endurance at 70 °C: advanced operation mode	$U = \sqrt{P_{70} \times \overline{R}}$ or $U = U_{\text{max.}}$; whichever is the less severe; 1.5 h on; 0.5 h off; 70 °C; 1000 h	$\pm (0.20\ \% R + 0.05\ \Omega)$	$\pm (0.50\ \% R + 0.05\ \Omega)$
7.3	-	Endurance at upper category temperature	125 °C; 1000 h 155 °C; 1000 h 175 °C; 1000 h	$\pm (0.05\ \% R + 0.01\ \Omega)$ $\pm (0.10\ \% R + 0.02\ \Omega)$ $\pm (0.25\ \% R + 0.05\ \Omega)$	$\pm (0.15\ \% R + 0.02\ \Omega)$ $\pm (0.30\ \% R + 0.02\ \Omega)$ $\pm (0.50\ \% R + 0.05\ \Omega)$
10.4	78 (Cab)	Damp heat, steady state	(40 $\pm$ 2) °C; (93 $\pm$ 3) % RH; 56 days	$\pm (0.10\ \% R + 0.01\ \Omega)$	$\pm (0.10\ \% R + 0.02\ \Omega)$
10.5	67 (Cy)	Damp heat, steady state accelerated: general operation mode	(85 $\pm$ 2) °C (85 $\pm$ 5) % RH $U = \sqrt{0.1 \times P_{70} \times \overline{R}}$; $U \leq 0.3 \times U_{\text{max.}}$; 1000 h whichever is the less severe	$\pm (0.10\ \% R + 0.05\ \Omega)$	$\pm (0.50\ \% R + 0.05\ \Omega)$



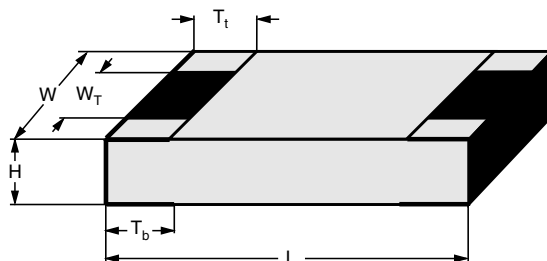
TEST PROCEDURES AND REQUIREMENTS					
EN 60115-1 CLAUSE	IEC 60068-2 ⁽¹⁾ TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR)	
				HIGH ACCURACY	REGULAR ACCURACY
			Stability for product types:		
			MCR 0201 AT	22 Ω to 40 k Ω	-
			MCS 0402 AT	1 Ω to $\leq$ 47 k Ω	> 47 k Ω to 1 M Ω
			MCT 0603 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 2 M Ω
			MCU 0805 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 7.5 M Ω
			MCA 1206 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 10 M Ω
10.4	2 (Bb) 30 (Db) 1 (Ab) 13 (M) 30 (Db) -	Climatic sequence: general operation mode		$\pm (0.25 \% R + 0.02 \Omega)$	
10.3.4.2		Dry heat	125 °C; 16 h		
10.3.4.3		Damp heat, cyclic	55 °C; 24 h; $\geq$ 90 % RH; 1 cycle		
10.3.4.4		Cold	-55 °C; 2 h		
10.3.4.5		Low air pressure	8.5 kPa; 2 h; (25 $\pm$ 10) °C		
10.3.4.6		Damp heat, cyclic	55 °C; 24 h; 90 % RH; 5 cycles		
10.3.4.7		DC load	$U = \sqrt{P_{70}} \times R \leq U_{\max.}$; 1 min		
-	1 (Ab)	Cold	-55 °C; 2 h	$\pm (0.05 \% R + 0.01 \Omega)$	
10.1	14 (Na)	Rapid change of temperature	30 min at -55 °C and 30 min at 155 °C; 1000 cycles	$\pm (0.10 \% R + 0.01 \Omega)$ R < 47R: $\pm (0.25 \% R + 0.05 \Omega)$	$\pm (0.25 \% R + 0.02 \Omega)$
		Extended rapid change of temperature	30 min at -40 °C; 30 min at 125 °C; MCS 0402 AT: 3000 cycles MCT 0603 AT: 2000 cycles MCU 0805 AT: 1500 cycles MCA 1206 AT: 1000 cycles	$\pm (0.25 \% R + 0.05 \Omega)$; ($\geq$ 50 % of initial shear force)	
8.1	-	Short time overload: general operation mode	$U = 2.5 \times \sqrt{P_{70}} \times R$ or $U = 2 \times U_{\max.}$; whichever is the less severe; 5 s	$\pm (0.05 \% R + 0.01 \Omega)$	
		Short time overload: power operation mode	$U = 2.5 \times \sqrt{P_{70}} \times R$ or $U = 2 \times U_{\max.}$; whichever is the less severe; 5 s	$\pm (0.1 \% R + 0.01 \Omega)$	$\pm (0.25 \% R + 0.05 \Omega)$
8.2	-	Single pulse high voltage overload: general operation mode	$U = 10 \times \sqrt{P_{70}} \times R$ or $U = 2 \times U_{\max.}$; whichever is the less severe; 10 pulses 10 μ s/700 μ s	$\pm (0.25 \% R + 0.05 \Omega)$	
		Single pulse high voltage overload: power operation mode	$U = 10 \times \sqrt{P_{70}} \times R$ or $U = 2 \times U_{\max.}$; whichever is the less severe; 10 pulses 10 μ s/700 μ s	$\pm (0.50 \% R + 0.05 \Omega)$	
8.4 ⁽²⁾	-	Periodic electric overload: general operation mode	$U = \sqrt{15 \times P_{70}} \times R$ or $U = 2 \times U_{\max.}$; whichever is the less severe; 0.1 s on; 2.5 s off; 1000 cycles	$\pm (0.50 \% R + 0.05 \Omega)$	
		Periodic electric overload: power operation mode	$U = \sqrt{15 \times P_{70}} \times R$ or $U = 2 \times U_{\max.}$; whichever is the less severe; 0.1 s on; 2.5 s off; 1000 cycles	$\pm (1.0 \% R + 0.05 \Omega)$	

TEST PROCEDURES AND REQUIREMENTS					
EN 60115-1 CLAUSE	IEC 60068-2 ⁽¹⁾ TEST METHOD	TEST	PROCEDURE	REQUIREMENTS PERMISSIBLE CHANGE (ΔR)	
				HIGH ACCURACY	REGULAR ACCURACY
			Stability for product types:		
			MCR 0201 AT	22 Ω to 40 k Ω	-
			MCS 0402 AT	1 Ω to $\leq$ 47 k Ω	> 47 k Ω to 1 M Ω
			MCT 0603 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 2 M Ω
			MCU 0805 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 7.5 M Ω
			MCA 1206 AT	1 Ω to $\leq$ 100 k Ω	> 100 k Ω to 10 M Ω
8.5	-	Electro static discharge (human body model)	IEC 61340-3-1; 3 pos. + 3 neg.; (MIL-STD-883, method 3015) MCR 0201 AT: 400 V MCS 0402 AT: 500 V MCT 0603 AT: 1000 V MCU 0805 AT: 1500 V MCA 1206 AT: 2000 V	$\pm (0.5 \% R + 0.05 \Omega)$	
9.11	6 (Fc)	Vibration	Endurance by sweeping; 10 Hz to 2000 Hz; no resonance; amplitude 1.5 mm or $\leq$ 200 m/s ² ; 7.5 h	$\pm (0.05 \% R + 0.01 \Omega)$ no visible damage	
11.1	58 (Td)	Solderability	Solder bath method; SnPb40; non-activated flux (215 $\pm$ 3) °C; (3 $\pm$ 0.3) s	Good tinning (95 % covered); no visible damage	
			Solder bath method; SnAg3Cu0.5 or SnAg3.5; non-activated flux; (235 $\pm$ 3) °C; (2 $\pm$ 0.2) s	Good tinning (95 % covered); no visible damage	
11.2	58 (Td)	Resistance to soldering heat	Solder bath method; (260 $\pm$ 5) °C; (10 $\pm$ 1) s ⁽³⁾	$\pm (0.05 \% R + 0.01 \Omega)$	$\pm (0.05 \% R + 0.01 \Omega)$ $R > 221 \text{ k}\Omega$ in 0402: $\pm (0.10 \% R + 0.01 \Omega)$ $R > 511 \text{ k}\Omega$ in 0603: $\pm (0.10 \% R + 0.01 \Omega)$ $R > 1 \text{ M}\Omega$ in 0805 / 1206: $\pm (0.10 \% R + 0.01 \Omega)$
11.3	45 (XA)	Component solvent resistance	Isopropyl alcohol +50 °C; method 2	No visible damage	
9.7	21 (Ue ₃)	Shear (adhesion)	MCR 0201 AT: 2 N	No visible damage	
			MCS 0402 AT and MCT 0603 AT: 9 N		
			MCU 0805 AT and MCA 1206 AT: 45 N		
9.8	21 (Ue ₁)	Substrate bending	Depth 2 mm, 3 times	$\pm (0.05 \% R + 0.01 \Omega)$ no visible damage; no open circuit in bent position	
12.2	-	Voltage proof	$U_{\text{RMS}} = U_{\text{ins}}$; (60 $\pm$ 5) s	No flashover or breakdown	
12.4	-	Flammability	IEC 60695-11-5, needle flame test; 10 s	No burning after 30 s	

Notes

- (1) The quoted IEC standards are also released as EN standards with the same number and identical contents
- (2) Periodic electric overload test for power operation mode not applicable for case size 0201
- (3) For MCR 0201 AT only similar to DIN EN 60115-8 test procedure. Due to the components small size they were fixed by glue previous to testing instead of applying tweezers

DIMENSIONS



DIMENSIONS AND MASS							
TYPE / SIZE	H (mm)	L (mm)	W (mm)	W _T (mm)	T _t (mm)	T _b (mm)	MASS (mg)
MCR 0201 AT	0.23 ± 0.03	0.6 ± 0.05	0.3 ± 0.05	> 50 % of W	0.12 ± 0.05	0.12 ± 0.05	0.14
MCS 0402 AT	0.32 ± 0.05	1.0 ± 0.05	0.5 ± 0.05	> 75 % of W	0.2 + 0.1 / - 0.15	0.2 ± 0.1	0.6
MCT 0603 AT	0.45 + 0.1 / - 0.05	1.55 ± 0.05	0.85 ± 0.1	> 75 % of W	0.3 + 0.15 / - 0.2	0.3 + 0.15 / - 0.2	1.9
MCU 0805 AT	0.52 ± 0.1	2.0 ± 0.1	1.25 ± 0.15	> 75 % of W	0.4 + 0.1 / - 0.2	0.4 + 0.1 / - 0.2	4.6
MCA 1206 AT	0.55 ± 0.1	3.2 + 0.1 / - 0.2	1.6 ± 0.15	> 75 % of W	0.5 ± 0.25	0.5 ± 0.25	9.2

SOLDERING RECOMMENDATIONS

For recommended solder pad dimensions please refer to www.vishay.com/doc?28950.

For recommended soldering profiles please refer to www.vishay.com/doc?31090.

REVISION HISTORY	
REVISION DATE	DESCRIPTION
27-Nov-2025	- Update of operating voltage rating for 0402 / 0603 - Improvement of test requirement
28-May-2026	- Added rated dissipation at terminal part temperature 105 °C - Update of material categorization - PFAS-free feature is valid for all products delivered with a date code of "2620" and later - Update of ESD voltage rating for 0201



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